

RAG Retrieval Methods and Technologies

Retrieval in RAG systems employs various techniques, each with distinct advantages and trade-offs. Lexical retrieval methods like BM25 use term frequency and document length normalization for efficient exact-match retrieval, performing well on structured queries but missing semantic similarity. Dense retrieval uses embedding models to convert text into vector representations, enabling semantic similarity search across high-dimensional spaces. Hybrid approaches combine lexical and dense retrieval, leveraging both exact matching and semantic understanding. Vector databases like FAISS, Pinecone, Weaviate, and Qdrant provide scalable storage and retrieval of embeddings. Approximate nearest neighbor (ANN) algorithms enable fast similarity search even with billions of embeddings. Graph-based retrieval methods organize documents hierarchically or through relationships, enabling more sophisticated reasoning. Multi-modal retrieval extends RAG to images, videos, and other media types by converting them to embeddings. Sparse retrieval methods like ColBERT create learned sparse vectors combining efficiency with semantic understanding. Cross-encoder rerankers further refine retrieved results by computing relevance scores for query-document pairs. The choice of retrieval method depends on corpus size, latency requirements, and specific use case needs.